

ADS-B Feature Importance Analysis

Which features actually flag which attack, verified against real labels

Bhuvan Krishna (Tony) · FL-IDS Cross-Domain Project · Data: opensky_5day_labeled.csv, 360,000-row stratified sample

Executive Summary

Unlike the UAV side of this project, ADS-B comes from a single, unified feature-engineering pipeline (not two mismatched capture rigs), so a genuine per-attack-type breakdown was possible here -- mutual information computed separately for each attack vs. benign, not just one blended multiclass score. Every number in this report is computed from the actual data, not domain inference.

Two findings stand out beyond the plain ranking. First, three currently-used features (is_stale, is_low_coverage_region, icao24_valid_format) carry essentially zero real signal -- is_stale and is_low_coverage_region measure exactly 0.0000 mutual information with attack label, meaning they contribute nothing to classification. Second, track_message_count shows a suspicious pattern: it's the #1 or #2 feature for four of five attack types, including gps_spoof and velocity_spoof, which don't actually modify this feature by design (only ghost_aircraft and flooding do). This is flagged in Section 3 as worth investigating rather than taken at face value.

1. Features Currently Used By The Final Model

The final reported model (trained on the original single-hour OpenSky file) used these 11 features. Note: callsign_valid_format was part of that original selection but no longer exists in this newer dataset -- the rewritten pipeline drops the raw callsign column entirely for memory, and no longer derives a validity flag from it. If retraining on this expanded dataset, that feature slot will need to be filled by something else.

Feature	Real rank (of 13)	MI vs. attack	RF importance	Verdict
inter_arrival_s	1	0.7491	0.2908	Strong
track_message_count	2	0.5614	0.2555	Strong, but see caveat
speed_consistency_delta	3	0.3209	0.0555	Strong
implied_speed_mps	4	0.3198	0.0609	Strong
vrate_consistency_delta	5	0.3129	0.0649	Strong
is_physically_implausible	6	0.1869	0.0470	Solid
turn_rate_deg_s	7	0.1769	0.0190	Solid
squawk_changed	10	0.0365	0.0533	Weak
icao24_valid_format	11	0.0136	0.0000	Very weak
is_stale	12	0.0000	0.0039	Zero signal
callsign_valid_format	n/a	n/a	n/a	No longer exists

2. How These Importances Were Calculated

Mutual information, computed TWO ways

An overall multiclass score (Section 4) shows each feature's general usefulness across all attack types blended together. A separate one-vs-benign score was computed independently FOR EACH attack type (Section 3) -- e.g. gps_spoof rows vs. benign rows only, ignoring every other attack type for that computation. This is what makes the per-attack answer real rather than inferred: each attack type's number reflects only that attack's actual effect on the data.

Random Forest importance (cross-check)

A Random Forest (200 trees, max depth 12, class-balanced, 84.7% multiclass test accuracy as a sanity check) was trained on the same data as a second, independent importance signal. Where it agrees with mutual information, that's real confirmation.

Why ADS-B doesn't need the UAV-style contamination check: ADS-B comes from one pipeline processing one consistent capture methodology, not two different capture rigs sharing only 47% of columns. There's no equivalent "which file did this come from" confound here.

3. How Features Correspond To Attacks (Verified, Not Inferred)

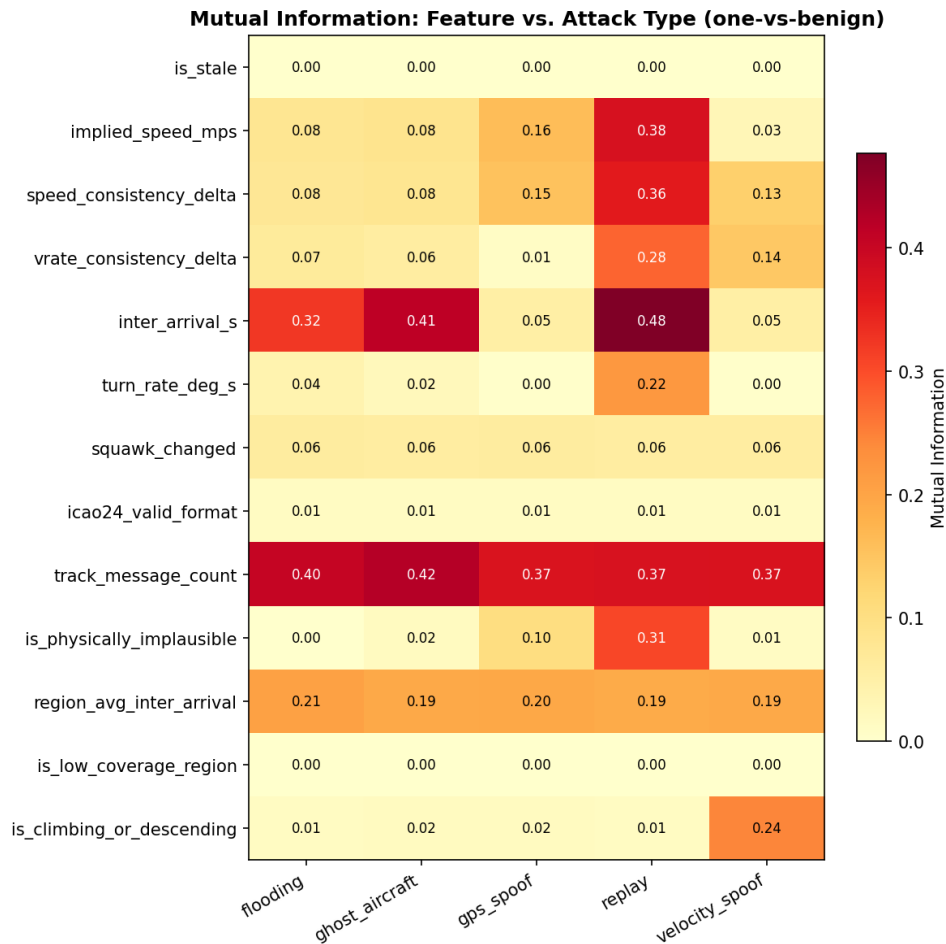


Figure 1. Mutual information between each feature and each specific attack type (one-vs-benign). Darker = stronger signal for that attack specifically.

Top 3 features per attack type

Attack	#1 feature	#2 feature	#3 feature
flooding	track_message_count (0.40)	inter_arrival_s (0.32)	region_avg_inter_arrival (0.21)
ghost_aircraft	track_message_count (0.42)	inter_arrival_s (0.41)	region_avg_inter_arrival (0.19)
gps_spoof	track_message_count (0.37)	region_avg_inter_arrival (0.20)	implied_speed_mps (0.16)
replay	inter_arrival_s (0.48)	implied_speed_mps (0.38)	track_message_count (0.37)
velocity_spoof	track_message_count (0.37)	is_climbing_or_descending (0.24)	region_avg_inter_arrival (0.19)

Worth investigating: `track_message_count` is the #1 feature for flooding, ghost_aircraft, and velocity_spoof, and #1-tied for gps_spoof -- but by the pipeline's own design, only ghost_aircraft and flooding actually modify this feature (see `apply_attack_to_subset` in the pipeline code). `gps_spoof` and `velocity_spoof` shouldn't touch it at all. Its uniformly high score (0.37-0.42) across ALL five attack types, rather than a sharp attack-specific spike, suggests either (a) a genuine real-world correlation between how often an aircraft is seen and its likelihood of being selected by the disjoint attack-assignment process, or (b) a subtle sampling artifact where attack assignment (proportional to row count) systematically favors aircraft that already have more rows. This is NOT confirmed either way here -- it's flagged as worth a direct follow-up check (e.g. compare `track_message_count` distributions between attack-assigned and benign rows for `gps_spoof` specifically) before leaning on this feature's apparent importance for attacks it wasn't designed to signal.

The genuinely clean, attack-specific signals are more informative for interpretation: `inter_arrival_s` spikes hardest for replay (0.48) and ghost_aircraft (0.41) -- both attacks that directly disrupt normal timing. `is_climbing_or_descending` spikes ONLY for velocity_spoof (0.24 vs. under 0.02 everywhere else) -- exactly matching the attack's actual mechanism (it fakes a climb rate). `turn_rate_deg_s` and `vrate_consistency_delta` both spike specifically for replay -- consistent with replay's timestamp-shifting mechanism distorting derived rate calculations.

4. Which Features Should Ideally Be Used

With only 13 real numeric candidates total (fewer than UAV's 64 raw columns), the honest recommendation is simple: drop the two features carrying zero measured signal, keep everything else.

Feature	MI vs. attack	Recommendation
<code>is_stale</code>	0.0000	DROP -- zero signal across every attack type and overall
<code>is_low_coverage_region</code>	0.0000	DROP -- zero signal across every attack type and overall
<code>icao24_valid_format</code>	0.0136	Weak but harmless -- low priority to replace
All other 10 features	0.15 - 0.75	KEEP -- all carry real, verified signal

Replacing `is_stale` and `is_low_coverage_region` frees two feature slots. The clearest candidates to fill them, both currently unused and both outranking several currently-used features: `is_climbing_or_descending` (rank 8, and the single cleanest attack-specific signal in the whole dataset for velocity_spoof) and `region_avg_inter_arrival` (rank 9, contributes real signal fairly evenly across all five attacks).

5. Overall Importance Chart

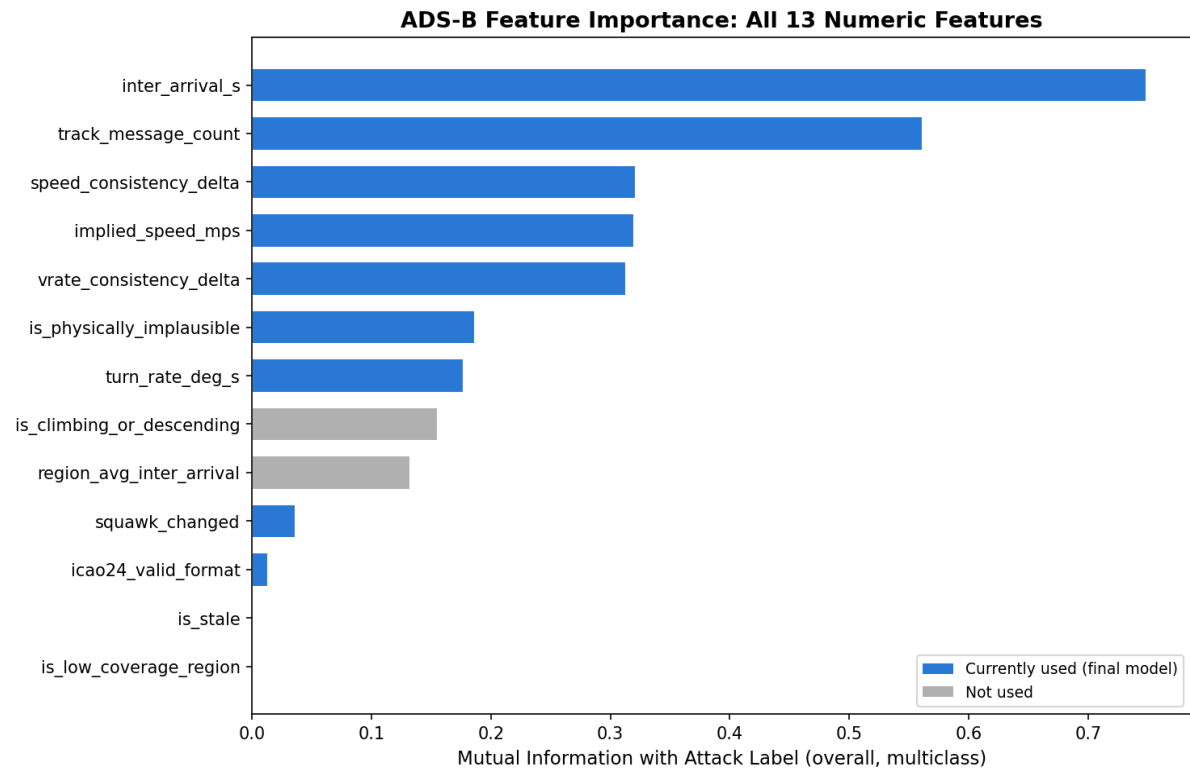


Figure 2. Overall (multiclass) mutual information for all 13 numeric features. Blue = used in the final reported model; grey = not used.

6. Full Table: All 13 Numeric Features

Feature	Rank	MI vs. attack (overall)	RF importance	Currently used?
inter_arrival_s	1	0.7491	0.2908	Yes
track_message_count	2	0.5614	0.2555	Yes
speed_consistency_delta	3	0.3209	0.0555	Yes
implied_speed_mps	4	0.3198	0.0609	Yes
vrate_consistency_delta	5	0.3129	0.0649	Yes
is_physically_implausible	6	0.1869	0.0470	Yes
turn_rate_deg_s	7	0.1769	0.0190	Yes
is_climbing_or_descending	8	0.1553	0.0745	No
region_avg_inter_arrival	9	0.1325	0.0738	No
squawk_changed	10	0.0365	0.0533	Yes
icao24_valid_format	11	0.0136	0.0000	Yes
is_stale	12	0.0000	0.0039	Yes
is_low_coverage_region	13	0.0000	0.0010	No

Limitations

- The track_message_count cross-attack dominance (Section 3) is flagged, not resolved -- worth a targeted follow-up before treating it as confirmed attack signal for gps_spoof/velocity_spoof specifically.
- callsign_valid_format was part of the original final model's feature set but no longer exists in the pipeline's current output -- any retraining on this dataset needs a replacement feature.
- Computed on a 360,000-row stratified sample (60,000 per class), not the full 4.6M-row file -- large-sample estimates would be more precise, though the clean class balance here (unlike UAV) makes this less of a concern than it was for the UAV analysis.
- Random Forest accuracy (84.7%) is a sanity check on the importance ranking, not a claim about the actual GAF/baseline models' real-world performance.